ZERO TRUST ACCESS DESIGN CLINIC · INTEGRATED PACKAGE

ZTA Activity Guide

A team design clinic · 4 roles · 5 use cases (3 foundational + 2 optional). Includes how your team will be scored.

The customer & the ask

The customer is a distributed enterprise: corporate HQ and on-prem data center, remote branches on SD-WAN, a hybrid workforce (some running AI agents), and applications across on-prem, AWS and Azure. Grown by acquisition, mid-flight into public cloud; security controls have not kept pace.

They are **not** asking you to rip anything out — bring Zero Trust to what already exists: **segment it, see it, and keep enforcing policy** wherever the user, device or workload is.

The environment

Network at a glance

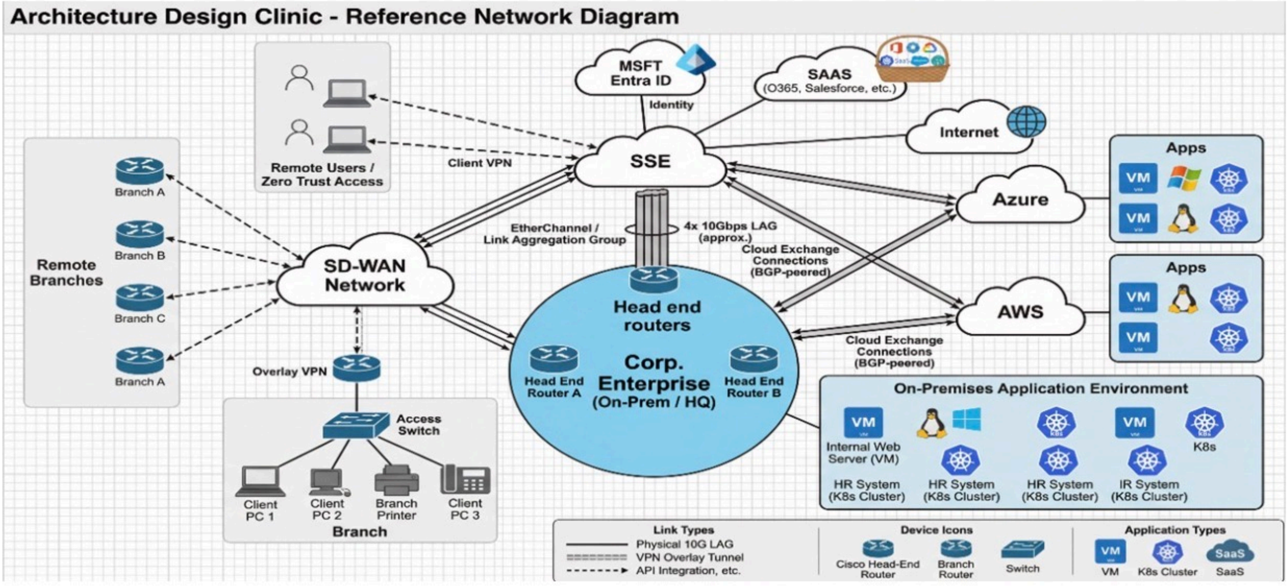
- **Campus HQ** — access/distribution/core switching, wired + wireless, large IoT footprint.
- **Branches** — over SD-WAN.
- **Data center** — on-prem apps + AWS/Azure workloads.
- **Wireless** — corporate + Guest SSID.
- **Identity/policy** — Cisco ISE.

Who's on the network

- **Employees & trusted devices** (Employees).
- **IoT** — printers, cameras, badge readers (IOT).
- **Deskside agents** — Mac-minis (deskagents).
- **Guests** — Internet-only (Guest).
- **Remote workers & contractors**; some running AI agents.

Reference architecture

Everyone starts from the same picture; your diagram is your version with the products and policy you propose on top.



Architecture Design Clinic - Reference Network Diagram
MSFT Entra ID
Identity
SAAS (O365, Salesforce, etc.)
Internet
Remote Users / Zero Trust Access
Client VPN
SSE
Remote Branches
Branch A
Branch B
Branch C
Branch A
SD-WAN Network
EtherChannel / Link Aggregation Group
4x 10Gbps LAG (approx.)
Cloud Exchange Connections (BGP-peered)
Azure
AWS
Apps
VM
Head end routers
Head End Router A
Corp. Enterprise (On-Prem / HQ)
Head End Router B
Overlay VPN
Access Switch
Client PC 1
Client PC 2
Branch Printer
Client PC 3
Branch
On-Premises Application Environment
Internal Web Server (VM)
K8s
HR System (K8s Cluster)
HR System (K8s Cluster)
HR System (K8s Cluster)
IR System (K8s Cluster)
Link Types
Physical 10G LAG
VPN Overlay Tunnel
API Integration, etc.
Device Icons
Cisco Head-End Router
Branch Router
Switch
Application Types
VM
K8s Cluster
SaaS

Four roles, no passengers

OWNS THE PACE

IT Director

Watches the clock and keeps the team moving; calls time checks; confirms each use case is submitted before moving on.

OWNS THE CHALLENGE

Digital Resiliency Officer

Challenges each decision ("what would make this fail?") to spark dialogue, surfaces the threat surface, makes the team justify trade-offs — not to block.

OWNS THE DESIGN

Network Architect

Holds the pen and draws the topology; ensures every product/control has a home on the diagram; decides placement and trust boundaries.

OWNS THE WRITE-UP

Network Security Engineer

Captures decisions; owns the pain → product mapping and product write-up; runs the diagram upload and final submission.

How the hour runs

- **Teams of 4** — one per role.
- **Five use cases — 1–3 foundational (required), 4–5 optional (if time permits)**; solve in order, submitting one unlocks the next.
- For each use case: map **pain → product**, list **products (how & why with Cisco differentiators)**, draw/expand a **topology diagram**, then submit for SME review and scoring.

How you'll be scored

Your proctor scores **each use case out of 100** across three criteria — **pain→product mapping (40)**, **products: how & why / differentiators (30)**, and **diagram & overall solution (30)**. The total places your solution at a **level**. Aim for **Expert**: name the right products, say *how* they work and *why* Cisco wins, and trace your policy on the diagram. (Full answer keys stay with the proctor.)

Associate `60–74`

Foundational: identifies the right general Cisco products and meets the basic requirement, but light on *how* and *why*.

Professional `75–89`

Solid, correct design: pains mapped to the right products with a clear how & why and a Cisco differentiator; enforcement points shown.

Expert `90–100`

Comprehensive & differentiated: every pain/requirement mapped precisely, multiple differentiators, flows traced end-to-end.

Below Associate = Developing (< 60): incomplete or largely incorrect.

What earns the marks

CRITERION	MAX	ASSOCIATE	PROFESSIONAL	EXPERT
Pain-point → product mapping	40	**20–27** Maps the main pain points to broadly correct products; some gaps or generic choices.	**28–34** Maps all pain points to the correct Cisco products, each with a clear reason.	**35–40** Every pain mapped precisely to the optimal product with strong justification; no gaps.
Products — how & why / differentiators	30	**15–20** Names products; explains "how" only partially; few or no differentiators.	**21–26** Correct "how" for each product and a real "why"; at least one Cisco differentiator.	**27–30** Precise "how" + compelling "why" for every product; multiple Cisco differentiators vs competitors.
Diagram & overall solution	30	**15–20** Diagram present and legible; labels groups; enforcement points partial or unclear.	**21–26** Labels groups, marks enforcement points; solution coherent and meets the requirements.	**27–30** Traces flow(s) end-to-end with policy/matrix detail; considers integration/edge cases; within budget.

Segmentation methodology

Shared vocabulary — reference, not the answer.

Why VLANs / subnets alone struggle

Identity-based (group) segmentation

- Segmentation tied to IP/VLAN means topology changes ripple everywhere.
- Access rules become long IP ACLs that are hard to read, audit and maintain.
- Every new device type needs new subnets and rules; staff can't keep up.

- Classify each user/device into a group at the point of access (authentication + profiling).
- Carry the group as a tag (SGT) with the traffic, independent of IP / VLAN / topology.
- Write policy as group-to-group, enforced across the existing network.

The building blocks

- Authenticate & profile every endpoint (802.1X, MAB, web auth).
- Posture — check endpoint health (AV, OS, patches) and remediate before full access.
- Tag & enforce — assign an SGT; switches / routers / firewalls enforce inline.
- Share context — feed identity/group via pxGrid to firewalls, DC, SD-WAN and threat tools.

How to approach each use case

- 1 · Map pain → product. For every pain point, name the product/feature that fixes it.
- 2 · Say how & why. One line on how it works and why it fits (differentiators welcome).
- 3 · Read the task & parameters. Note the specific groups, access rules and constraints.
- 4 · Propose a diagram. Draw the topology with your solution on it, then submit.

The use cases

Use Case 1 — Managed Environment — Segmentation

Required

The customer's policy is to segment campus & branch wired and wireless client devices (laptops, mobile devices) into logical groups.

Customer requirements

- Employee and trusted devices (Group=Employees).
- IOT — printers, cameras, badge readers, etc. (Group=IOT).
- Deskside agents — dedicated mac-minis (Group=deskagents).
- Guest SSID for Internet only (Group=Guest).
- IOT should be restricted from the other groups but need data center and Internet access.
- Guests should have Internet access only.
- Employee and IOT require controlled internet, and data center application access.
- Deskside Mac-Minis have access to on-prem data center only.

Customer pain points

- Prior segmentation failed. Using VLANS/IP subnets is complex and tedious.
- Profiling is labor intensive and staff unable to keep up with constant new devices joining.
- Failing PCI audits.
- Corporate PCs not running latest AntiVirus and OS updates.
- No budget for buying more firewalls to do segmentation.

Group tasks

- Draw a diagram of the customer's campus and branch wired/wireless networks. Submit for SME review and scoring.
- Create your proposed segmentation solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your solution addresses each customer requirement and pain point.

Use Case 2 — Remote Workers — Zero Trust Access

Required

Customer looking for a SASE solution to support remote workers and secure branch connectivity.

Customer requirements

- All corporate users can use full client VPN to access selected legacy private applications. But most prefer ZTNA for modern applications.
- All corporate users need controlled internet access (block gambling and gaming).
- Contractors can only access private applications via clientless RDP access.
- Improve user experience with minimal application sign-ons.
- Customer wants a SASE solution for branches.

Customer pain points

- Full client VPN has over-permissive privileges. Not considered Zero Trust.
- User satisfaction is low due to manual VPN connection daily.
- Too many sign-ons per application. Low user satisfaction.
- Want to eliminate password logins to minimize stolen credentials.
- Existing Okta IdP solution getting too expensive.
- CIO wants SASE to reduce branch costs.

Group tasks

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your solution addresses each customer requirement and pain point.

Use Case 3 — AI Access and Agent Controls

Required

The customer is concerned about uncontrolled AI usage by users from Use Case 2. They want to report what's being used and build enforceable guardrails on what AI tools can do — controlling which AI tools are safe and validating information shared with approved AI sites.

Customer requirements

- User agents should not be granted rights to use tools equally. Fine-grained control is needed.
- Corporate users should only be able to login to their organization's ChatGPT tenant.
- Some remote users will be running Claude Desktop agents. Ensure your design includes controls for those employee systems.
- Some users will be integrating MCP servers to local agents like Claude Code.

Customer pain points

- Shadow AI usage is an issue today.
- Too many unknown MCP servers for applications in both the private and public DCs.
- Users using personal credentials for non-enterprise ChatGPT login.
- Downloading of risky AI models.

Group tasks

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed agentic solution based on what was taught in the agentic AI design presentation, and be ready to explain the product selected, how it works, and why (include Cisco differentiators). Make sure your solution addresses each customer requirement and pain point.

Use Case 4 — Merger & Acquisition — Extend Segmentation

Bonus

The customer recently acquired a small company (~100 staff) and needs to integrate it into the network as quickly as possible. The remote location has a Cisco FTD as its Internet edge and another segmenting the local AI cluster; the acquired company uses Okta as its IdP.

Customer requirements

- CISO wants to extend segmentation to the acquired company using TrustSec.
- Only selected branch users can access a restricted resource at the recently acquired company (Site 9951).

Customer pain points

- Unsure how to extend segmentation to the acquired company.
- Parent company is not ready to add 100 newly acquired staff into their corporate directory. Is there a quick option of using what they already have?
- During transition, minimize disruptive user authentication experience to corporate apps and quickly secure the acquired company apps / user population.
- Acquired company suffered an identity breach previously. Still using passwords.

Group tasks

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your solution addresses each customer requirement and pain point.

Use Case 5 — Zero Trust across Multicloud Data Center

Bonus

The customer is challenged with application sprawl across on-prem, AWS and Azure data centers. Propose a strategy and design to identify their applications (including Kubernetes) and minimize the blast radius. A comprehensive segmentation solution including application run-time protection is highly desired.

Customer requirements

- Limited budget. Want to leverage native security features when practical. Open to virtual appliances.
- CIO seeking an end-to-end segmentation solution.
- Minimize complexity.

Customer pain points

- Minimal visibility of applications across multicloud data centers.
- Lacking segmentation strategy and plan.
- Wide deployment of Kubernetes with latency and visibility issues.
- Unable to patch IOT devices.
- Too many disparate security management interfaces for on-prem and multicloud enforcement points.

Group tasks

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your solution addresses each customer requirement and pain point.